

TestLab 29119 — User Manual

Course: SENG 436 — Learner-as-Designer Project
Institution: Çankaya University
Standards: ISO/IEC 29119-4 (Test Techniques) · ISO/IEC 25010 (Quality Model)
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1. Introduction

TestLab 29119 is a browser-based educational serious game. You play the role of a **Test Manager** who must classify, apply, and select ISO/IEC 29119-4 test design techniques for realistic software projects. The game reinforces correct technique choice—not memorization of definitions alone—through four progressive phases and immediate feedback.

2. System Requirements

Requirement	Details
Browser	Recent Chrome, Microsoft Edge, or Firefox
JavaScript	Must be enabled
Internet	Required for Google Fonts (optional fallback to system fonts)
Storage	<code>localStorage</code> used for best score, leaderboard, and theme
Installation	None (web application)

3. How to Access the Game

Online (recommended for submission)

Live URL: <https://ismaiidogan.github.io/testlab-game/>

Local (development or offline demo)

```
npm install
npm start
```

Open `http://localhost:3000` in your browser.

4. Game Overview

Item	Description
Maximum score	500 points
Phases	4 + Results
Duration	Approximately 25–40 minutes (first playthrough)
Language	English

Phase summary

Phase	Name	Max points	Skill tested
1	Classify	110	Map techniques to Specification / Structure / Experience categories
2	Apply	195	Choose techniques for nine realistic scenarios
3	Strategize	110	Build a technique mix for three projects (Clause 4 constraints)
4	Test Cases	85	Select required EP and BVA test cases
—	Results	—	Grade, misconceptions, certificate, leaderboard

5. Getting Started

5.1 Welcome screen

1. Optionally enter **Your Name** and **School Name** (used on the leaderboard and certificate).
2. Click **Start Mission** to open the tutorial.
3. Defaults: Anonymous / Unknown School if fields are left empty.

5.2 Tutorial (six slides)

Slide	Topic
1	Welcome — Test Manager role
2	ISO/IEC 29119-4 — three technique categories
3	The three categories (Specification / Structure / Experience)
4	Clause 4 selection criteria
5	ISO/IEC 25010 link to Phase 2 scenarios

Slide	Topic
6	Your mission — overview of all four phases

- Use **Next** / **Back** and the dot indicator to navigate.
- Click **Skip** to go directly to Phase 1.
- On the last slide, **Start Phase 1** begins the game.

6. Phase-by-Phase Instructions

Phase 1 — Classify the Techniques

Goal: Drag each of **15 technique cards** into the correct category drop zone.

Zone	Category
Blue	Specification-Based
Green	Structure-Based
Red	Experience-Based

Actions:

- **Drag and drop** cards from the pool into a zone.
- **i** button — opens ISO detail modal (clause, description, best use).
- **Hint** button — reveals a hint; costs **4 points** per first use on each card.
- When all cards are placed, click **Check Answers**.
- Correct placements are highlighted in green; wrong cards show feedback with the correct category.
- Click **Continue to Phase 2** when finished.

Scoring: 6 points per correct card; +20 bonus if all 15 are correct; hint penalties subtracted from Phase 1 total (max 110).

Phase 2 — Apply to Scenarios

Goal: Answer **9 multiple-choice scenarios** by choosing the best test technique(s).

Features:

- Each scenario shows context, a question, and four options (A–D).
- **Keyboard:** Press **A**, **B**, **C**, or **D** to select; **Escape** closes modals.
- **ISO/IEC 25010 badge** on the scenario header links quality characteristics to the scenario.
- Some scenarios are **comparison mode** (e.g. EP vs BVA, Exploratory vs Ad Hoc).

- After answering, read **correct/wrong feedback** and any **misconception** note.
- Click **Next Scenario** until all nine are complete.

Scoring: 20–25 points per scenario (max 195 for Phase 2).

Phase 3 — Build Your Test Strategy

Goal: For each of **three projects**, select up to **five** test techniques that best fit the constraints.

Projects:

1. MedSafe — Healthcare (safety-critical)
2. QuickMVP — Startup food delivery (time/budget limited)
3. MigrateX — Legacy ERP migration (undocumented rules)

Actions:

- Read **constraints** (risk, budget, timeline, team, documentation).
- Click techniques to select/deselect (maximum 5).
- Hover for a quick preview; use the glossary (book icon) for ISO terms.
- Click **Submit Strategy** to see score, optimal selection, expert analysis, and key insight.
- Click **Next Project** or **Continue to Phase 4** on the last project.

Scoring: Up to 37 / 37 / 36 points per project (max 110). Partial credit applies; there is no single fixed algorithm—matching Clause 4 professional judgment.

Phase 4 — Write Your Test Cases

Goal: For **two exercises**, select all **required** EP and BVA test cases (checkboxes).

Exercises:

1. Login form — username field (5–20 characters)
2. Age field — insurance discount (18–65 inclusive)

Actions:

- Click rows to toggle selection.
- Click **Submit** to see coverage feedback (correct, missed, or unnecessary selections).
- Extra (non-required) selections incur a small penalty.
- Click **Next Exercise** or **See Final Results** on the last exercise.

Scoring: Up to 42 and 43 points (max 85). Each required item = 6 points when selected.

7. Scoring and Grades

Total score

Your **total** is the sum of all four phase scores (max **500**). The navbar **Score** updates as you play; **Best** stores your highest total in this browser.

Grade tiers (percentage of 500)

Grade	Minimum %	Title
ISO Master	90%	Outstanding mastery
Senior Tester	75%	Strong understanding
Test Engineer	60%	Good basics
Junior Tester	40%	Learning in progress
Trainee	0%	Review recommended

8. User Interface Reference

Navbar (top)

Control	Function
Top 10	Local leaderboard (this browser only)
Best	Highest total score saved locally
Score	Current run total
Book	ISO reference glossary (searchable)
Sun/Moon	Light / dark theme
Speaker	Mute / unmute sound effects

Phase progress bar

Shows four steps: **Classify** → **Apply** → **Strategize** → **Test Cases**. Active and completed steps are highlighted during gameplay.

9. Results Screen

After Phase 4 you will see:

- **Grade title** and emoji (e.g. ISO Master)
- **Total points** and percentage
- **Animated progress bar**
- **Per-phase score breakdown**
- **Misconceptions encountered** (learning points from wrong answers)
- **Phase 3 key insights**
- **Hint usage** summary (Phase 1)
- **Download Certificate** — PNG file with your name, grade, and scores
- **Play Again** — returns to welcome (scores reset for a new run)

10. Troubleshooting

Problem	Solution
Score or leaderboard wrong	Data is per-browser; clear site data or use another browser
Fonts look plain	Check internet connection (Google Fonts CDN)
Drag-and-drop not working	Use a desktop browser; refresh the page
Game does not load	Ensure JavaScript is enabled; use the GitHub Pages URL or <code>npm start</code>

11. Appendix A — Screenshot Walkthrough

Item 3 of the course deliverable list applies to physical non-web games. As a web application, TestLab 29119 is accessed via URL; the following ordered screenshots document the user interface.

#	File	Screen
1	01-welcome.png	Welcome and player profile
2	02-tutorial-categories.png	Tutorial — three technique categories
3	03-tutorial-clause4.png	Tutorial — Clause 4 selection criteria
4	04-phase1-classify.png	Phase 1 — drag-and-drop classification
5	06-phase2-scenario.png	Phase 2 — scenario with ISO 25010 badge
6	07-phase3-strategy.png	Phase 3 — project and technique selection
7	08-phase4-testcases.png	Phase 4 — EP/BVA test case selection
8	09-results.png	Results — score and grade breakdown
9	10-glossary.png	ISO reference glossary panel

Screenshots are stored in the `docs/screenshots/` folder of the project repository.

12. Appendix B — Regenerating Screenshots (developers)

```
npx playwright test tests/capture-screenshots.spec.js
```

End of User Manual